Practice Problems
Calculus-Based Physics II

Name: Key **Date:** _____

Chapter 35 – Electrostatic Force

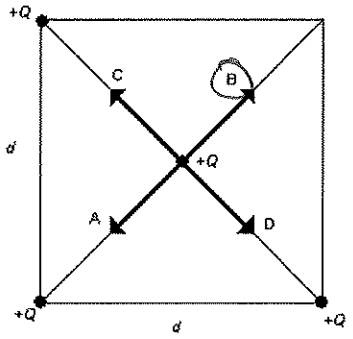
1) Two tiny beads are 25 cm apart with no other charges or fields present. Bead A carries 10 μC of charge and bead B carries 1 μC. Which one of the following statements is true about the magnitudes of the electric forces on these beads?

$|\vec{F}_{A1}| = \frac{k q_1 q_2}{r^2}$ $|\vec{F}_{B1}| = \frac{k q_2 q_1}{r^2}$ } Same magnitude

A) The force on A is 10 times the force on B.
B) The force on B is 10 times the force on A.
C) The force on A is exactly equal to the force on B.
D) The force on A is 100 times the force on B.
E) The force on B is 100 times the force on A.

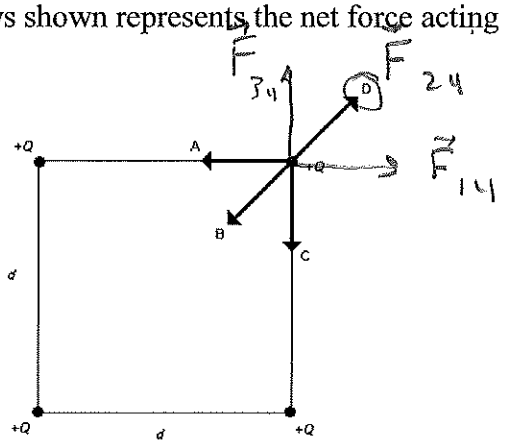
2) Three equal positive charges Q are at three of the corners of a square of side d. A fourth positive charge Q of equal magnitude is at the center of the square as shown in the figure Which one of the arrows shown represents the net force acting on the charge at the center of the square?

A) A
B) B
C) C
D) D



3) Four point charges of equal magnitude and charge (positive) are arranged on three of the corners of the square of side d as shown in the figure below. Which of the arrows shown represents the net force acting on the charge at the upper right-hand corner of the square?

A) A
B) B
C) C
D) D
E) none of the above



4) An electron and a proton are released simultaneously from rest and start moving toward each other due to their electrostatic attraction, with no other forces present. Which of the following statements are true just before they are about to collide?

A) They are closer to the to the initial position of the electron than they are to the initial position of the proton.
(B) They are closer to the to the initial position of the proton than they are to the initial position of the electron.
C) They are at the midpoint of their initial separation.
D) They both have the same speed.
E) The electrostatic force on the proton is greater than the electrostatic force on the electron.

$m_{e^-} << m_{p^+}$

5) Particle A has twice the charge of nearby particle B. Compared to the force on Particle A, the force on Particle B is

A) half as much.
B) four times as much.
C) twice as much.
(D) the same.
E) None of the above choices are correct.

$|\vec{F}_{AB}| = \frac{kq(2q)}{r^2}$ ↓ equal $|\vec{F}_{BA}| = \frac{k(2q)q}{r^2}$

6) Two objects with equal net charges of magnitude 1 C and positive are separated by a distance of 1 km. What is the force acting between these two objects? $k = 9 \times 10^9 \, Nm^2/C$

$q_1 = 1C$
$q_2 = 1C$
$r = 1 \times 10^3 m$

$|\vec{F}| = \frac{kq_1q_2}{r^2} = 9{,}000N$

7) The force of attraction between an object with a net negative charge of 40.0 μC and a net positive charge of 108 μC charge is 4.00 N. What is the separation between these two charges?

$|\vec{F}| = \frac{kq_1q_2}{r^2}$

Solve For "r"

$r^2 = \frac{kq_1q_2}{|\vec{F}|}$

$r = \sqrt{\frac{kq_1q_2}{|\vec{F}|}} = 3.12m$

8) Two 10¢ coins (dimes) having the same net charges are lying 2.5 m apart on a table. If each of these coins experiences an electrostatic force of magnitude 2.0 N due to the other coin, how large is the charge on each coin? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N · m^2/C^2)

$|\vec{F}_E| = \frac{kqq}{r^2} = \frac{kq^2}{r^2}$

Solve For q

$q^2 = \frac{|\vec{F}|r^2}{k}$

$q = \sqrt{\frac{|\vec{F}|r^2}{k}} = \pm 3.7 \times 10^{-5} C$

9) Two <u>equally charged</u> tiny spheres of mass 1.0 g are placed 2.0 cm apart. When released, they begin to accelerate away from each other at 414 m/s^2. What is the magnitude of the charge on each sphere, assuming only that the electric force is present? ($k = 1/4\pi\varepsilon_0 = 9.0 \times 10^9$ N · m^2/C^2)

$\vec{F}_E$

$\Sigma F_x = F_E = ma_x$

$F_E = \frac{kqq}{r^2}$

$\frac{kq^2}{r^2} = ma_x$

Solve for q

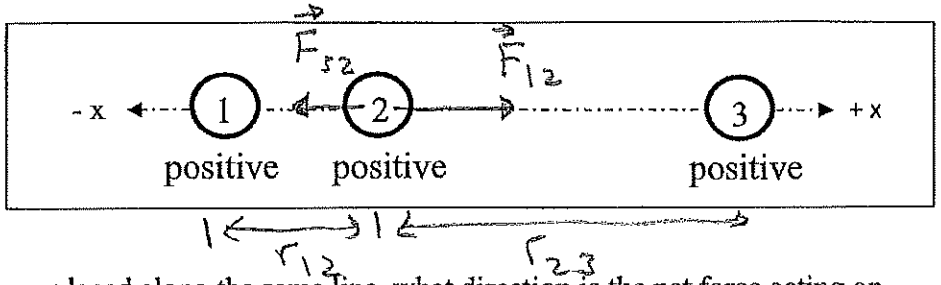
$q^2 = \frac{ma_x r^2}{k}$

$q = \sqrt{\frac{ma_x r^2}{k}}$

$q = 1.36 \times 10^{-7}$ C

10) Three spheres of equal charge are placed along the same line, what direction is the net force acting on sphere 2?

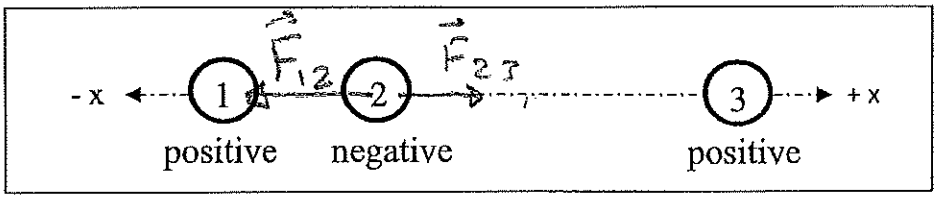
A) Positive x direction
B) Negative x direction
C) Net force is zero



$|\vec{F}_{12}| > |\vec{F}_{32}|$

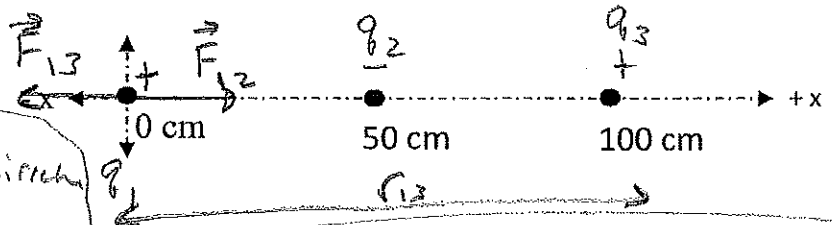
11) Three spheres of equal charge are placed along the same line, what direction is the net force acting on sphere 2

A) Positive x direction
B) Negative x direction
C) Net force is zero



$|\vec{F}_{12}| > |\vec{F}_{23}|$

12) Three point charges are placed on the x-axis. A positive charge of 2.0 μC is placed at the origin, a negative 2.0 μC to the right at $x = 50$ cm, and a positive 4.0 μC at the 100 cm mark. What are the magnitude and direction of the electrostatic force which acts on the charge at the origin?



$\Sigma \vec{F} = \vec{F}_{12} + \vec{F}_{13}$

$\Sigma F_x = |\vec{F}_{12}| + (-|\vec{F}_{13}|)$ ← - x direction

$|\vec{F}_{12}| = \frac{kq_1q_2}{r_{12}^2} = 0.144$ N

$|\vec{F}_{13}| = \frac{kq_1q_3}{r_{13}^2} = 0.072$ N

$\Sigma F_x = 0.144\text{ N} + (-0.072\text{ N})$

$\Sigma F_x = +0.072$ N

+ x-direction

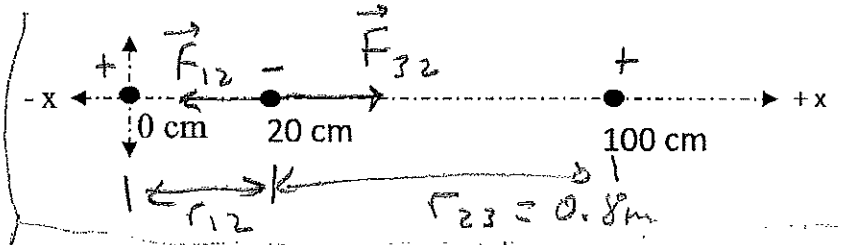
13) Three point charges are placed on the x-axis. A positive charge of 2.0 μC is placed at the origin, a negative 2.0 μC to the right at $x = 20$ cm, and a positive 4.0 μC at the 100 cm mark. What are the magnitude and direction of the electrostatic force which acts on the charge at the 20 cm position?

$\Sigma \vec{F} = \vec{F}_{32} + \vec{F}_{12}$

$\Sigma F_x = |\vec{F}_{32}| + (-|\vec{F}_{12}|)$

$\Sigma F_x = 0.1125N - 0.9N$

$\Sigma F_x = -0.79N$



$|\vec{F}_{12}| = \frac{kq_1q_2}{r_{12}^2} = 0.9N$ $\quad |\vec{F}_{32}| = \frac{kq_2q_3}{r_{32}^2} = 0.1125N$

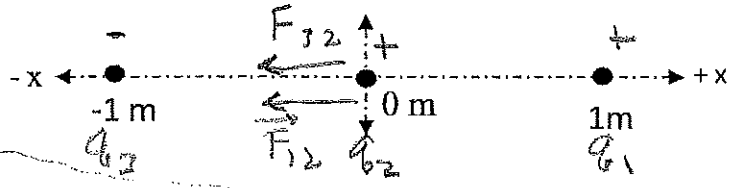
14) Three point charges are located on the x-axis at the following positions: a positive $Q_1 = 2.00$ μC is at $x =$ 1.00 m, a positive $Q_2 = 3.00$ μC is at x = 0.00, and a negative $Q_3 = 5.00$ μC is at $x = -1.00$ m. What is the magnitude of the electric force on $Q2$? ($k = 1/4\pi\varepsilon_0 = 8.99 \times 10^9$ N · m^2/C^2)

$\Sigma \vec{F} = \vec{F}_{32} + \vec{F}_{12}$

$\Sigma F_x = (-|\vec{F}_{32}|) + (-|\vec{F}_{12}|)$

$\Sigma F_x = -0.135N + (-0.054N)$

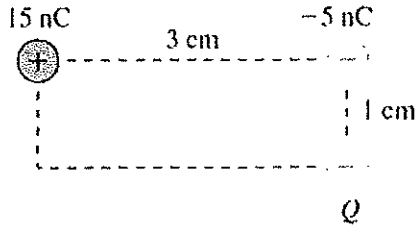
$\Sigma F_x = -0.189N$



$|\vec{F}_{32}| = \frac{kq_3q_2}{r_{32}^2} = 0.135N$

$|\vec{F}_{12}| = \frac{kq_1q_2}{r_{12}^2} = 0.054N$

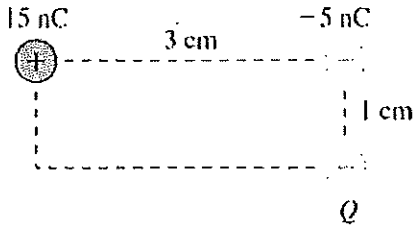
15) As shown in the figure, three charged objects are placed at corners of a rectangle. A 15 nC positive charged object at the upper left, a negative 5 nC object at the top right. The negative charge in the bottom right corner is $Q = 90$ nC. What is the magnitude of the net electrical force on the negative 5 nC charge due to the other two charges? ($k = 9.0 \times 10^9$ N · m^2/C^2)



See attached

$|\Sigma \vec{F}| = 0.0405N$

16) As shown in the figure, three charged objects are placed at corners of a rectangle. A 15 nC positive charged object at the upper left, a negative 5 nC object at the top right. The negative charge in the bottom right corner is $Q = 90$ nC. What is the magnitude of the net electrical force on the 90 nC charge Q due to the other two charges? ($k = 9.0 \times 10^9$ N · m^2/C^2)

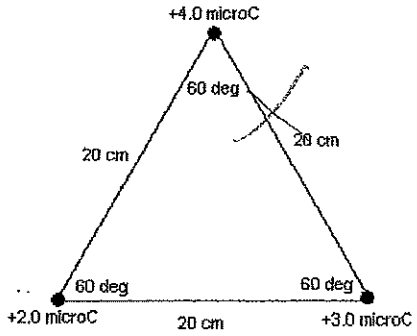


$|\Sigma \vec{F}| =$ 0.038N

See attached

17) Three object with a net charge are located at the corners of a triangle as shown below, a positive charge of 2.0 μC, a positive charge of 3.0 μC, positive charge of 4.0 μC. Determine the magnitude of the resultant force acting on the positive 4.0 μC charge.

See attached

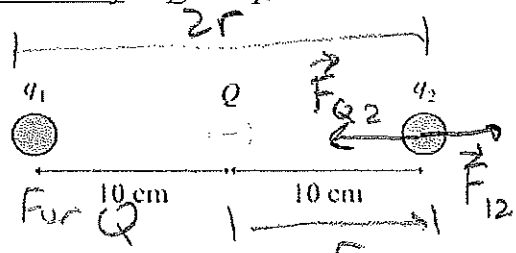


18) As shown in the figure, the charge Q is midway between two other charges. If Q = 7.5 nC and is negative, what must the net charge q_1 be so that charge q_2 remains stationary as Q and q_1 are held in place?

$\Sigma \vec{F} = \vec{F}_{Q_2} + \vec{F}_{12} = 0N$

$|F_{Q2}| = \frac{kQq_2}{r^2}$ $|\vec{F}_{12}| = \frac{kq_1q_2}{(2r)^2}$

$\Sigma \vec{F} = -\frac{kQq_2}{r^2} + \frac{kq_1q_2}{(2r)^2} = 0$



Solve For Q

$Q = \frac{r^2 k q_1 q_2}{4 r^2 k q_2} = \frac{q_1}{4}$

$q_1 = 4Q$

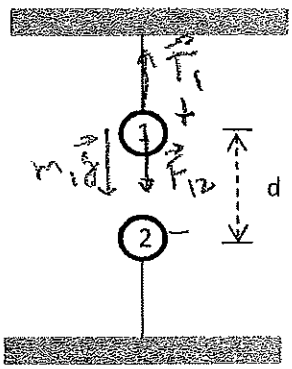
19) The figure below shows two spheres attached to a string, sphere 1 has a mass of 0.2 kg, a net positive charge of q1 = 5 μC and is attached to the ceiling, sphere 2 has a mass of 0.1 kg and a net negative charge of 5 μC and is attached to the floor. If the two sphere are separated by d = 0.3 m, what is the tension in the string holding sphere 1?

$\Sigma \vec{F} = \vec{T}_1 + m_1\vec{g} + \vec{F}_{12} = 0N$

$\Sigma F_y = T_1 + (-m_1 g) + (-|\vec{F}_{12}|) = 0N$

$T_1 = m_1 g + |\vec{F}_{12}| = 1.96N + 2.5N$

$T_1 = 4.46N$



$|\vec{F}_{12}| = \frac{kq_1q_2}{d^2} = 2.5N$

$m_1 g = 1.96N$

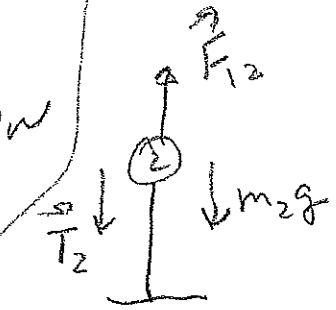
20) The figure above shows two spheres attached to a string, sphere 1 has a mass of 0.2 kg, a net positive charge of q1 = 5 μC and is attached to the ceiling, sphere 2 has a mass of 0.1 kg and a net negative charge of 5 μC and is attached to the floor. If the two sphere are separated by d = 0.3 m, what is the tension in the string holding sphere 2?

$\Sigma \vec{F}_2 = \vec{F}_{12} + \vec{T}_2 + m_2\vec{g} = 0N$

$\Sigma F_y = |\vec{F}_{12}| + (-T_2) + (-m_2 g) = 0N$

$T_2 = |\vec{F}_{12}| - m_2 g = 2.5N - 0.98N$

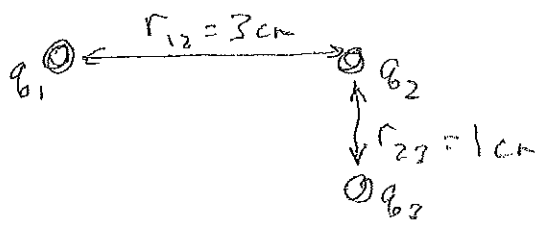
$T_2 = 1.52N$



$|\vec{F}_{12}| = \frac{kq_1q_2}{d^2} = 2.5N$

$m_2 g = 0.98N$

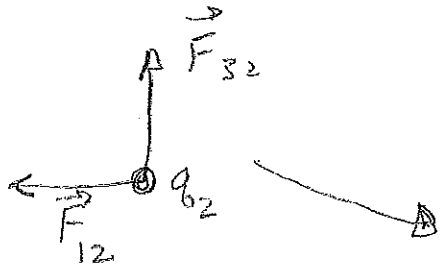
15)



$q_1 = 15\,nC$ positive
$q_2 = 5\,nC$ negative
$q_3 = 90\,nC$ negative

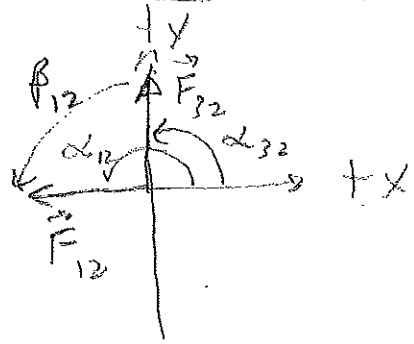
$r_{12} = 3\,cm = 3\times10^{-2}\,m$
$r_{23} = 1\,cm = 1\times10^{-2}\,m$

For q_2



$$\Sigma \vec{F}_{q_2} = \vec{F}_{32} + \vec{F}_{12}$$

need in vector component Form



$$\Sigma \vec{F}_{q_2} \cdot \hat{x} = \vec{F}_{32}\cdot\hat{x} + \vec{F}_{12}\cdot\hat{x} = \Sigma F_x$$

$$\Sigma F_x = |\vec{F}_{32}|\cos(\alpha_{32}) + |\vec{F}_{12}|\cos(\alpha_{12})$$

$$\Sigma F_x = 0 + |\vec{F}_{12}|(-1)$$

$\beta_{32} = 0°$, $\beta_{12} = 90°$ } From +y axis

$$\Sigma \vec{F}_{q_2}\cdot\hat{y} = \vec{F}_{32}\cdot\hat{y} + \vec{F}_{12}\cdot\hat{y} = \Sigma F_y$$

$$\Sigma F_y = |\vec{F}_{32}|\cos(\beta_{32}) + |\vec{F}_{12}|\cos(\beta_{12})$$

$$\Sigma F_y = |\vec{F}_{32}|(1) + 0$$

$$\Sigma \vec{F}_{q_2} = (\Sigma F_x)\hat{x} + (\Sigma F_y)\hat{y}$$

$$\Sigma \vec{F}_{q_2} = (-|\vec{F}_{12}|)\hat{x} + (|\vec{F}_{32}|)\hat{y}$$

magnitudes

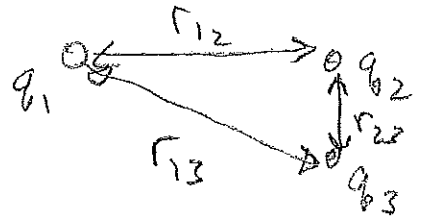
$$|\vec{F}_{32}| = \frac{k q_2 q_3}{r_{32}^2} = 0.0405\,N$$

$$|\vec{F}_{12}| = \frac{k q_1 q_2}{r_{12}^2} = 7.5\times10^{-4}\,N$$

$$|\Sigma \vec{F}_{q_2}| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2}$$

$$|\Sigma \vec{F}_{q_2}| = 0.0405\,N$$

16)



$r_{12} = 3\,cm$

$r_{23} = 1\,cm$

$r_{13} = \sqrt{r_{12}^2 + r_{23}^2}$

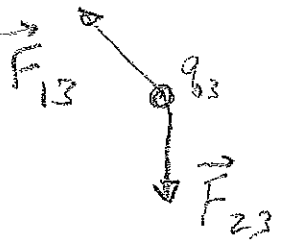
$r_{13} = 3.16\,cm$

$q_1 = 15nC$ positive

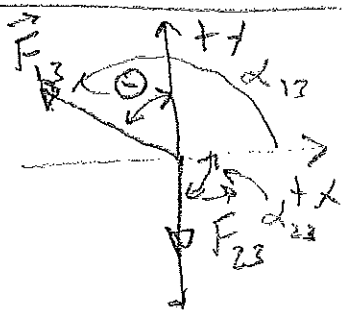
$q_2 = 5nC$ negative

$q_3 = 90nC$ negative

For q_{63}



$\Sigma \vec{F}_{q_3} = \vec{F}_{13} + \vec{F}_{23}$



need vector components

$\Sigma \vec{F}_{q_3} \cdot \hat{x} = \vec{F}_{13} \cdot \hat{x} + \vec{F}_{23} \cdot \hat{x} = \Sigma F_x$

$\Sigma F_x = |\vec{F}_{13}| \cos(\alpha_{13}) + |\vec{F}_{23}| \cos(\alpha_{23})$

$\Sigma F_x = |\vec{F}_{13}| \cos(\theta + 90^\circ)$

$\tan\theta = \frac{3cm}{1cm}$

$\theta = \tan^{-1}(3)$

$\theta = 71.56^\circ$

$\Sigma \vec{F}_{q_3} \cdot \hat{y} = \vec{F}_{13} \cdot \hat{y} + \vec{F}_{23} \cdot \hat{y} = \Sigma F_y$

$\Sigma F_y = |\vec{F}_{13}| \cos(\beta_{13}) + |\vec{F}_{23}| \cos(\beta_{23})$

$\beta_{13} = \theta$

$\beta_{23} = 180^\circ$

$\Sigma F_y = |\vec{F}_{13}| \cos(\theta) + (F_{23})(-1)$

$\Sigma \vec{F}_{q_3} = (|F_{13}|\cos(\theta + 90^\circ))\hat{x} + (|F_{13}|\cos(\theta) - |F_{23}|)\hat{y}$

$\Sigma \vec{F}_{q_3} = (-0.0114N)\hat{x} + (0.0038N - 0.0405N)\hat{y}$

$\Sigma \vec{F}_{q_3} = (-0.0114N)\hat{x} + (-0.0367N)\hat{y}$

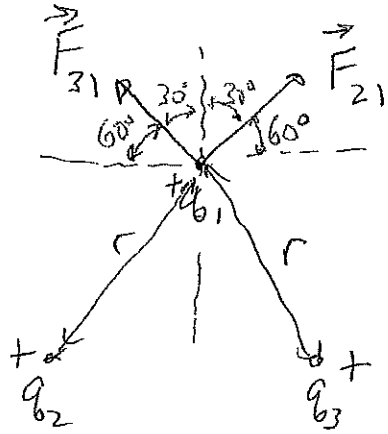
magnitudes

$|\vec{F}_{13}| = \frac{K q_1 q_3}{r_{13}^2} = 0.012N$

$|\vec{F}_{23}| = \frac{K q_2 q_3}{r_{23}^2} = 0.0405N$

$|\Sigma \vec{F}_{q_3}| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 0.038N$

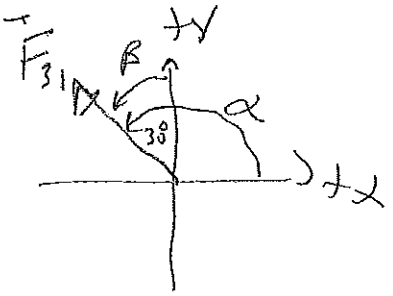
17)



$$|\vec{F}_{31}| = \frac{k q_3 q_1}{r^2} = 2.7N$$

$$|\vec{F}_{21}| = \frac{k q_2 q_1}{r^2} = 1.8N$$

$$\Sigma \vec{F} = \vec{F}_{31} + \vec{F}_{21}$$



$$\vec{F}_{31} = (|\vec{F}_{31}|\cos(\alpha))\hat{x} + (|\vec{F}_{31}|\cos(\beta))\hat{y}$$

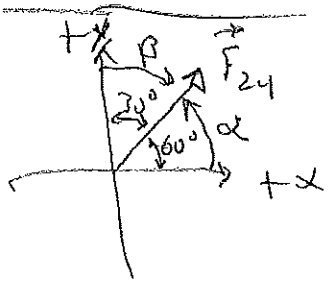
$\alpha = 90° + 30° = 120°$ $\quad \beta = 30°$

$$\vec{F}_{31} = (-1.35)\hat{x} + (+2.34N)\hat{y}$$

$$\vec{F}_{21} = (|\vec{F}_{21}|\cos(\alpha))\hat{x} + (|\vec{F}_{21}|\cos(\beta))\hat{y}$$

$\alpha = 60°$ $\quad \beta = 30°$

$$\vec{F}_{21} = (+0.9N)\hat{x} + (+1.56N)\hat{y}$$



$$\Sigma \vec{F} = (\Sigma F_x)\hat{x} + (\Sigma F_y)\hat{y} = (-1.35N + 0.9N)\hat{x} + (2.3N + 1.56N)\hat{y}$$

$$\Sigma \vec{F} = (-0.45N)\hat{x} + (3.86N)\hat{y}$$

$$|\Sigma \vec{F}| = \sqrt{(\Sigma F_x)^2 + (\Sigma F_y)^2} = 3.89N$$

$$\alpha = \cos^{-1}\left(\frac{\Sigma F_x}{|\Sigma \vec{F}|}\right) = 96.6°$$

$$\beta = \cos^{-1}\left(\frac{\Sigma F_y}{|\Sigma \vec{F}|}\right) = 7.1°$$